

Protein-source tryptophan as an efficacious treatment for social anxiety disorder: a pilot study.

Until recently, intact protein that is rich in tryptophan was not seen as an alternative to pharmaceutical-grade tryptophan because protein also contains large neutral amino acids (LNAAs) that compete for transport sites across the blood-brain barrier. Recent evidence indicates that when deoiled gourd seed (a rich source of tryptophan with approximately 22 mg/g protein) is combined with glucose (a carbohydrate that reduces serum levels of competing LNAAs) a clinical effect similar to that of pharmaceutical-grade tryptophan is achieved. Objective and subjective measures of anxiety in those suffering from social phobia (also known as social anxiety disorder) were employed to measure changes in anxiety in response to a stimulus as part of a double-blind, placebo-controlled, crossover study with a wash-out period of 1 week between study sessions. Subjects were randomly assigned to start with either (i) protein-source tryptophan (deoiled gourd seed) in combination with carbohydrate or (ii) carbohydrate alone. One week after the initial session, subjects returned for a follow-up session and received the opposite treatment of that received at the first session. All 7 subjects who began the study completed the 2-week protocol. Protein-source tryptophan with carbohydrate, but not carbohydrate alone, resulted in significant improvement on an objective measure of anxiety. Protein-source tryptophan combined with a high glycemic carbohydrate is a potential anxiolytic to those suffering from social phobia.